

# Hassan Javaid Rao

AI Engineer · Senior Software Engineer — LLM Applications, RAG Systems & Voice AI

Multan, Pakistan • hassanjrao@gmail.com • +92 303 520 9409 • hassanrao.com • LinkedIn • GitHub

## PROFESSIONAL SUMMARY

AI-focused Senior Software Engineer with **5+ years of professional experience**, including 4+ years building GCC enterprise logistics systems processing thousands of shipments daily across Oman, UAE, and Saudi Arabia. Specialized in **applied LLM engineering**: RAG pipelines, grounded AI assistants, and real-time AI voice agents. Initiated and lead the AI program at one of Oman's largest logistics platforms — shipping RAG chatbots and AI copilots used daily by operations teams — and independently built a **multi-tenant AI call-agent SaaS** that answers live phone calls with hallucination-resistant, knowledge-grounded responses. Strong foundation in scalable AWS architecture, multi-tenant SaaS, and technical mentorship. Section Leader at Stanford University's Code in Place program.

## EXPERIENCE

### Thiqa Tech — Muscat, Oman

Dec 2021 – Present

*Promoted from Software Engineer to Senior Software Engineer; 5 years of continuous ownership of the same multi-tenant logistics SaaS platform.*

#### ▸ Senior Software Engineer

Jan 2024 – Present

- **AI initiative** Proposed, architected, and lead the platform's AI program — taking LLM features from concept to production for one of Oman's largest logistics companies (15 branches).
- Built **RAG-based customer chatbots for shippers**: embedding + vector-retrieval pipelines over tenant shipment data and policy documents, with grounding rules that keep answers accurate, scoped, and hallucination-free.
- Shipped **"Ask AI" for operations** — a natural-language RAG assistant that answers any question about shipments, replacing minutes of manual cross-system lookups with instant grounded answers.
- Automated shipment data entry with **Google Gemini**, eliminating ~6 hours of daily manual effort and materially reducing operating cost.
- Architected and led the **migration from a single monolith to a modular architecture** — defined module boundaries and drove incremental extraction with zero downtime.
- Built an internal real-time chat system used across branches (NestJS, MongoDB, Node.js, Next.js).
- Engineered a high-performance export pipeline on **AWS Lambda + SQS** that reliably exports millions of records at minimal cost.
- Mentor junior engineers through code reviews and architectural guidance; maintain 99.9% uptime with Sentry and CloudWatch observability.

#### ▸ Software Engineer

Dec 2021 – Dec 2023

- Core engineer on a **multi-tenant logistics SaaS built from zero**, scaling it to thousands of shipments per day and millions of records across Oman, UAE, and Saudi Arabia.
- Designed the platform's **role-based access control (RBAC)** spanning tenants, branches, and operational roles.
- Built secure REST APIs with **OAuth 2.0** — Password Grant for first-party mobile apps, Client Credentials for machine-to-machine integrations with logistics partners **DHL, AJEX, and iMile**.
- Improved system performance ~20% by refactoring legacy code into modern patterns; introduced Redis-backed queues to keep UX smooth under heavy load.
- Stack: Laravel, Vue.js, Node.js, MySQL, Redis, AWS.

### Freelance Full-Stack Developer — Fiverr & Upwork

2020 – 2021

- Top Rated on both platforms with **250+ five-star reviews**; delivered web and automation solutions for 200+ clients across logistics, e-commerce, and SaaS.

- Competitively selected to teach Python to international learners in Stanford's flagship open-access program; led weekly sessions and gave personalized feedback.

## PROJECTS

### AI Caller — Multi-tenant AI Voice-Agent SaaS (founder/solo build)

[ai-caller.hassanrao.com](https://ai-caller.hassanrao.com)

Python (FastAPI), Laravel, Vue.js, PostgreSQL + pgvector, Deepgram Voice Agent, Twilio, Google Gemini, AWS

- Built and launched a SaaS where any business self-serves an **AI phone receptionist** that answers real inbound calls 24/7 in a natural voice — live in production on AWS.
- Engineered the **real-time voice pipeline**: Twilio Media Streams bridged to Deepgram's agent API (STT → LLM → TTS) over WebSockets, with barge-in interruption handling and **LLM function calling** for shipment tracking, quote capture, and human transfer.
- Designed the **RAG grounding layer**: PDF knowledge bases chunked and embedded (Gemini) into pgvector with HNSW cosine indexes; tenant-scoped retrieval with relevance thresholds forces honest "I don't know" responses instead of hallucinations, with prompt-injection defenses around retrieved content.
- **LLM-automated onboarding**: when a company signs up in a new industry, Gemini researches the domain and writes that industry's agent rulebook (scope fences, compliance rules) automatically.
- Full SaaS product: self-serve signup with plans, per-tenant agent configuration (voice, model, prompts), bring-your-own-Twilio with API-verified credentials and automatic webhook setup, and complete call transcripts with captured leads.
- Deployed on AWS EC2 (nginx, systemd) with GitHub Actions CI/CD across both services.

### Translation Management API

PHP 8.2, Laravel, MySQL, Redis, Docker

- Production-grade REST API for locale-based translations: Controller → Service → Repository architecture, Redis tag-based cache invalidation, and cursor-based export streaming millions of records at low memory cost; fully Dockerized with a PHPUnit suite covering API contracts, caching, and exports.

## TECHNICAL SKILLS

|                                 |                                                                                                                                                                                                                        |
|---------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>AI / LLM Engineering</b>     | RAG pipelines, embeddings & vector search (pgvector), prompt engineering & grounding, LLM function/tool calling, hallucination mitigation, voice agents (Deepgram, Twilio), Google Gemini APIs, AI workflow automation |
| <b>Backend</b>                  | Python (FastAPI), PHP (Laravel), Node.js (NestJS), TypeScript                                                                                                                                                          |
| <b>Frontend</b>                 | Vue.js, React.js / Next.js, Inertia.js, Tailwind CSS                                                                                                                                                                   |
| <b>Cloud &amp; DevOps</b>       | AWS (Lambda, SQS, S3, EC2, CloudWatch), Docker, GitHub Actions CI/CD, nginx, Redis                                                                                                                                     |
| <b>Databases</b>                | PostgreSQL (+ pgvector), MySQL, MongoDB, Redis                                                                                                                                                                         |
| <b>Architecture</b>             | Multi-tenant SaaS, modular monolith & microservices, event-driven systems, REST API design, OAuth 2.0 / JWT                                                                                                            |
| <b>Quality &amp; Monitoring</b> | Sentry, AWS CloudWatch, PHPUnit, Git, Agile/Scrum                                                                                                                                                                      |

## EDUCATION

### B.S. Software Engineering — Sukkur IBA University

2017 – 2020

Fully funded merit-cum-need scholarship (4 years) • IELTS 7.0 • Coursework: AI, Data Structures & Algorithms, OOP, Database Systems, Operating Systems, Web Engineering

## AWARDS

Winner — Web Design Competition, Sukkur IBA • Winner — Code in the Dark, Sukkur IBA • 4-year fully funded merit scholarship